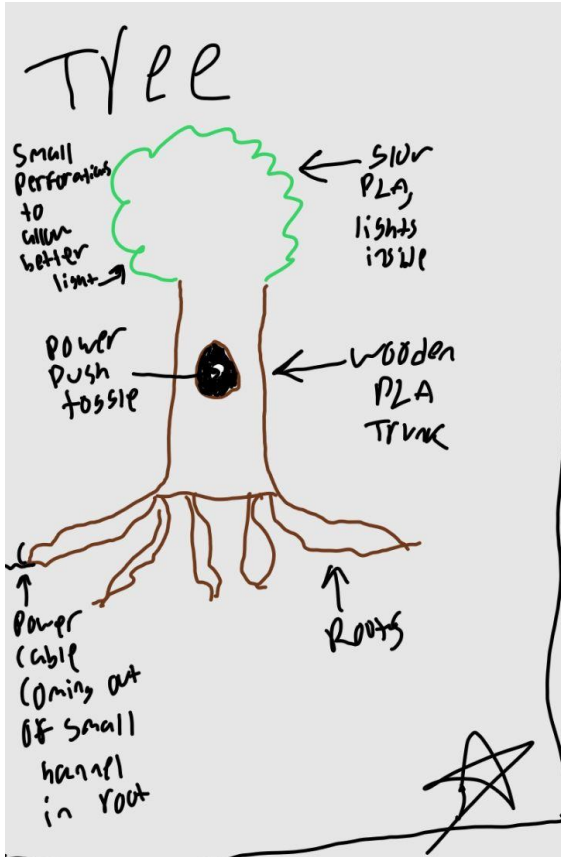
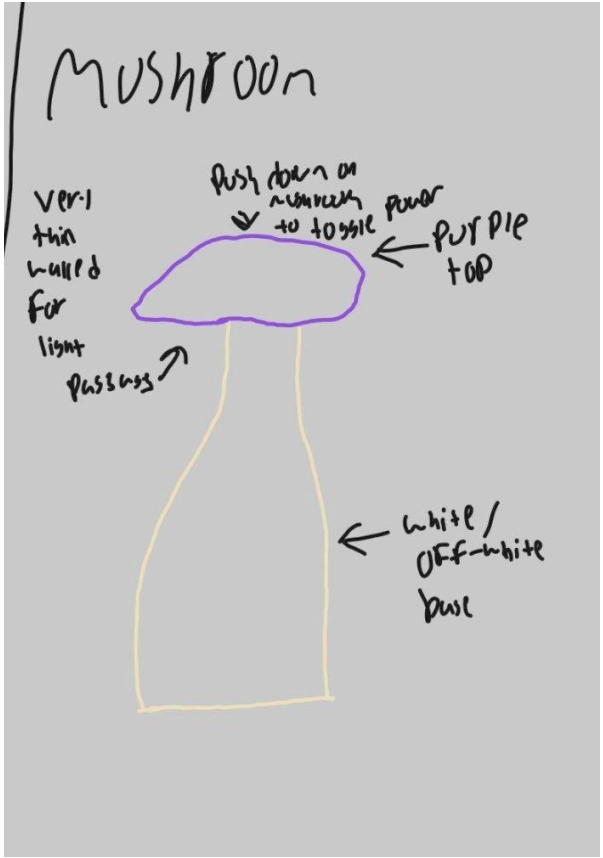
Lamp Portfolio

Ayan Bindal

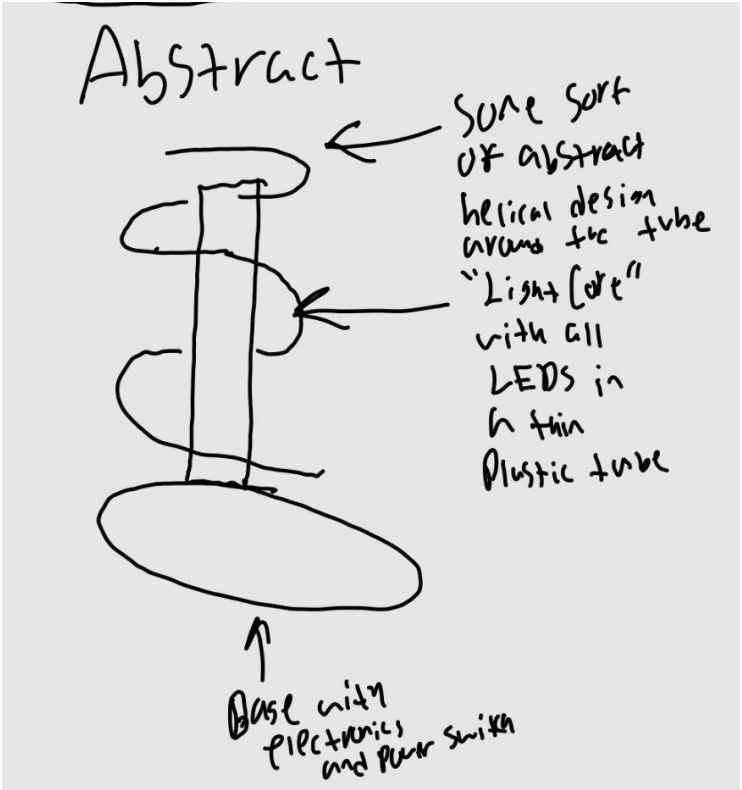
Idea #1 - Tree


Tree
Small perforations to allow better light
Silver PLA, lights inside
Power push fossil
Wooden PLA Trunk
Power cable coming out of small hole in root
Roots

Idea #2 - Mushroom



Idea #3 - Abstract



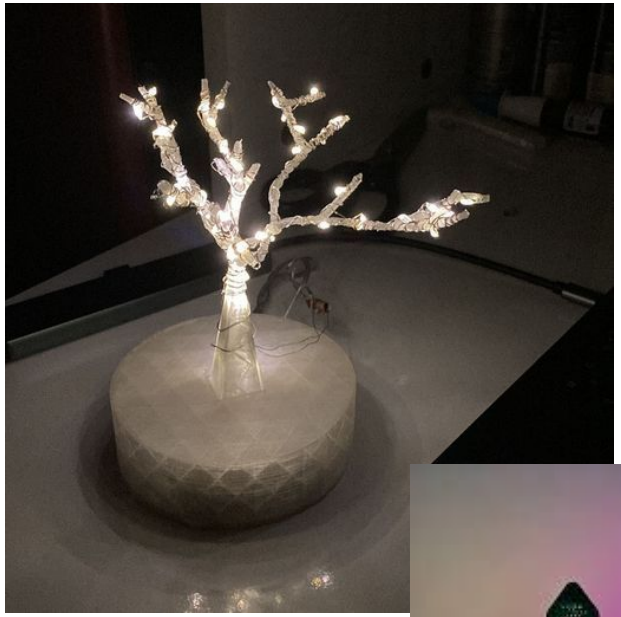
Goals

A realistic tree lamp that when turned off, glows in the dark and becomes a decorative piece, but when on becomes a source of bright, natural light.

Why?

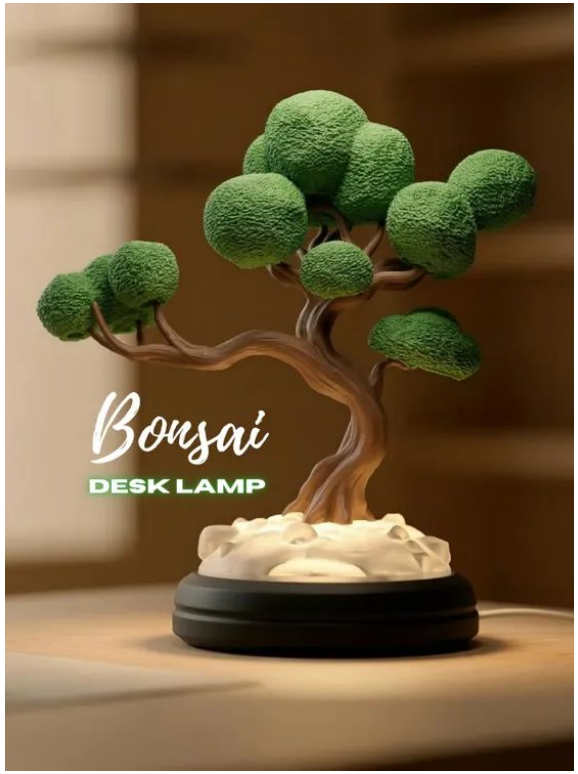
I thought that a lamp that doesn’t look like a lamp at first glance would be really cool, and the glow-in-the-dark feature sounds really nice.

Inspiration Pictures





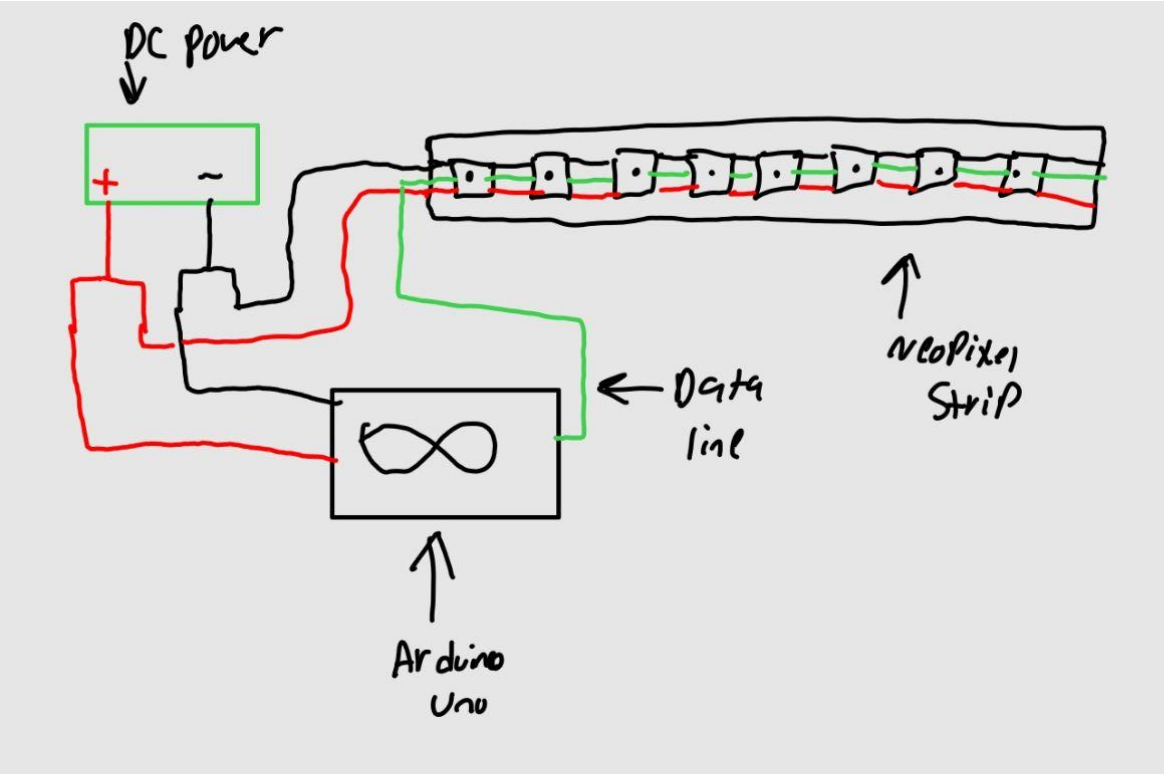




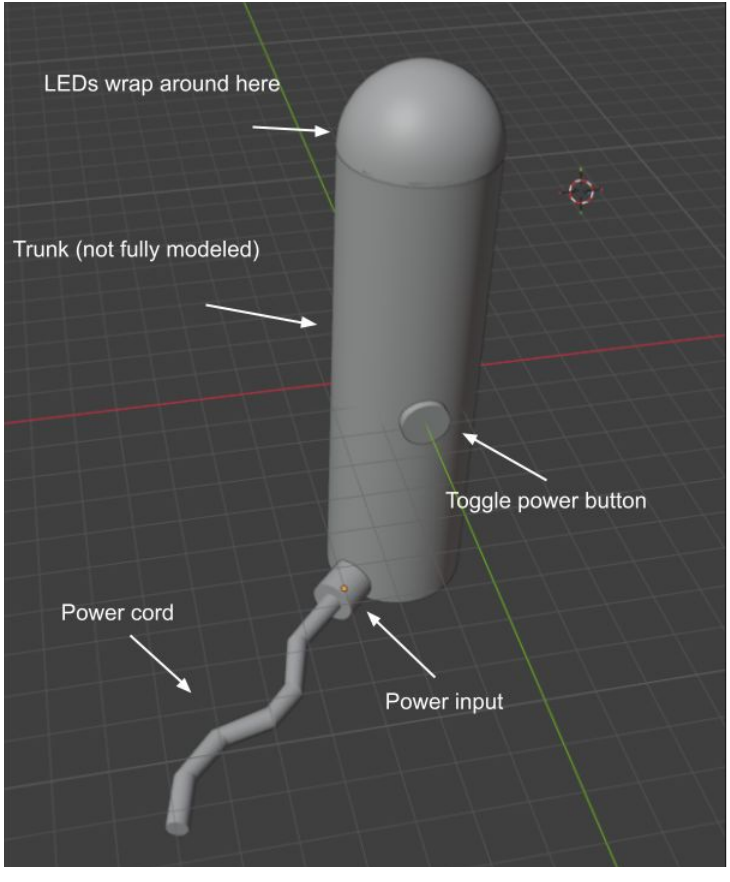
Materials to be used

- Wood PLA for trunk
- Glow PLA for leaves
- Push toggle switch for power
- Arduino UNO
- Neopixel strip (thicker one with white flex PCB) - 3ft length
- DC to screw terminal block
- DC power adapter (5V/2A)

Circuit



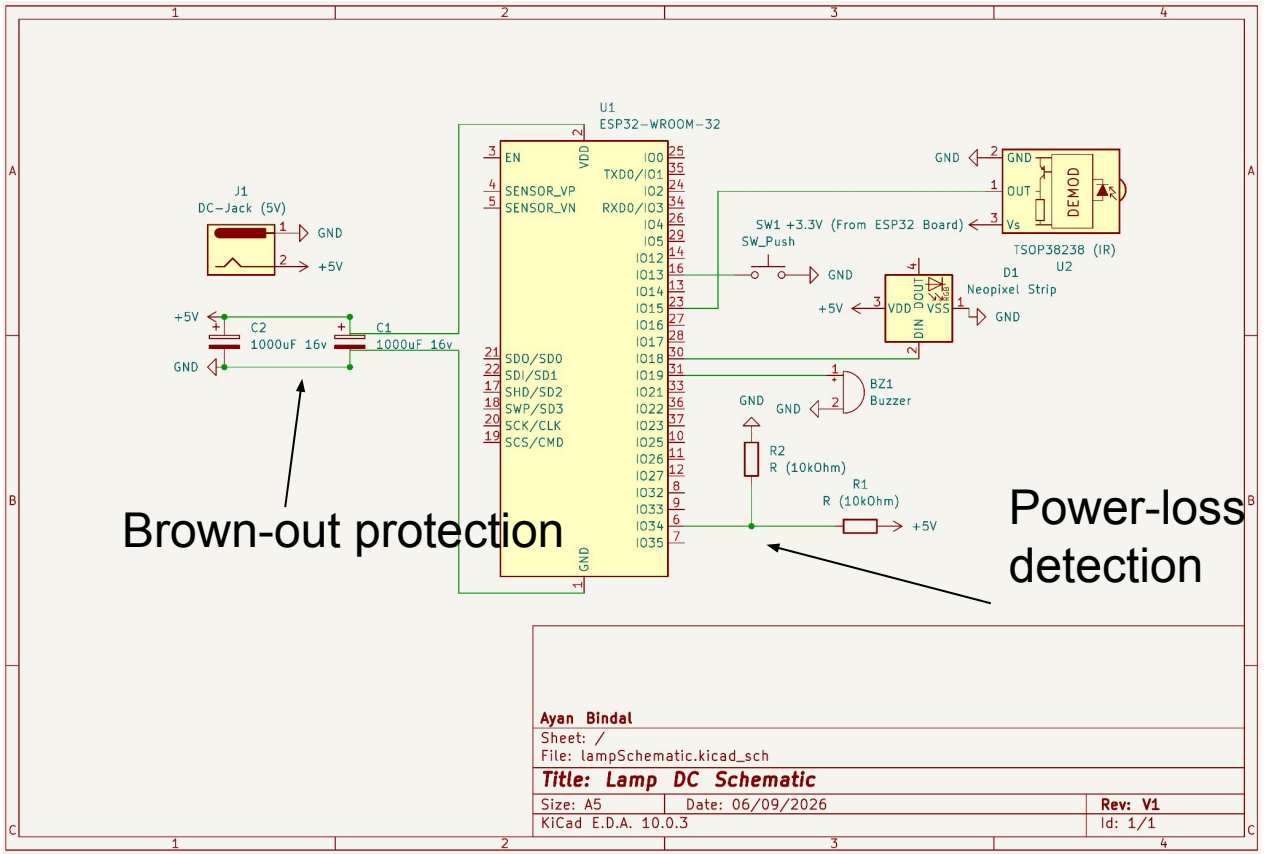
Initial Model (Internals)



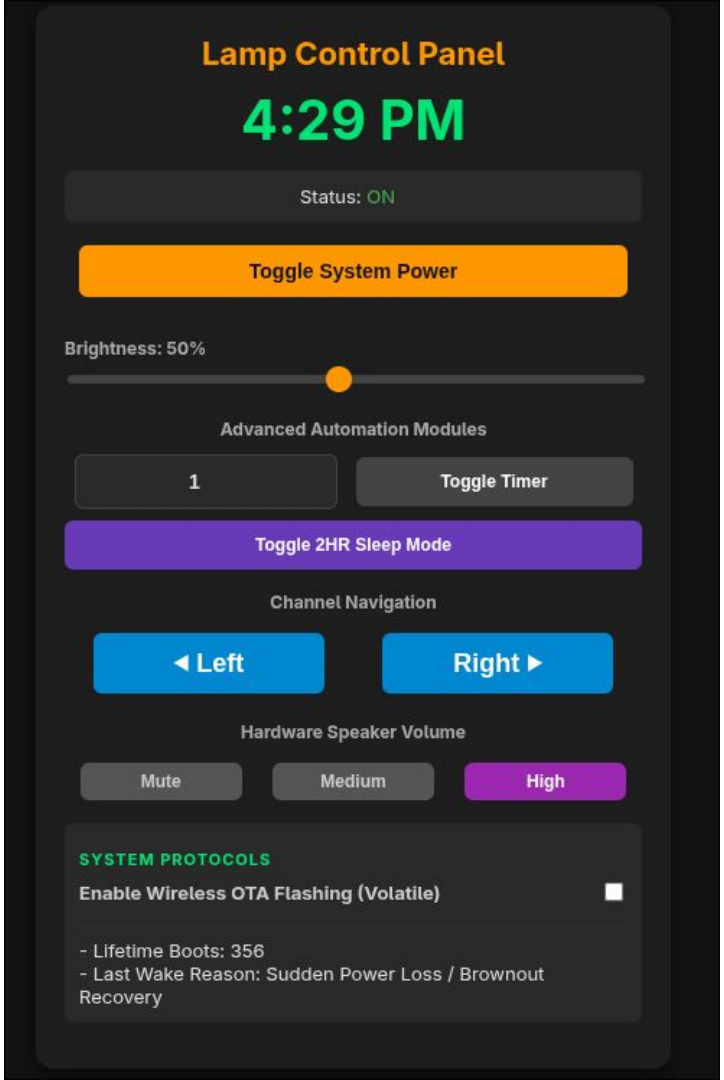
Design Update

- Added IR receiver (*Enables remote control*)
- Switched to ESP32 (*Allows for accurate time (sunset/sunrise timers) and wireless control*)
- Switched to swappable external shell (the visible tree part) (*Allows for designs to be easily swapped by swapping the shell rather than transplanting electronics*)
- Added various features through WebUI (Timer, night light, wireless updating)
- Added a buzzer (*To beep the IP address of the lamp & general sound effects*)
- Added power-loss recovery & brown-out protection (*Remembers lamp state by saving it to flash before capacitors drain*)

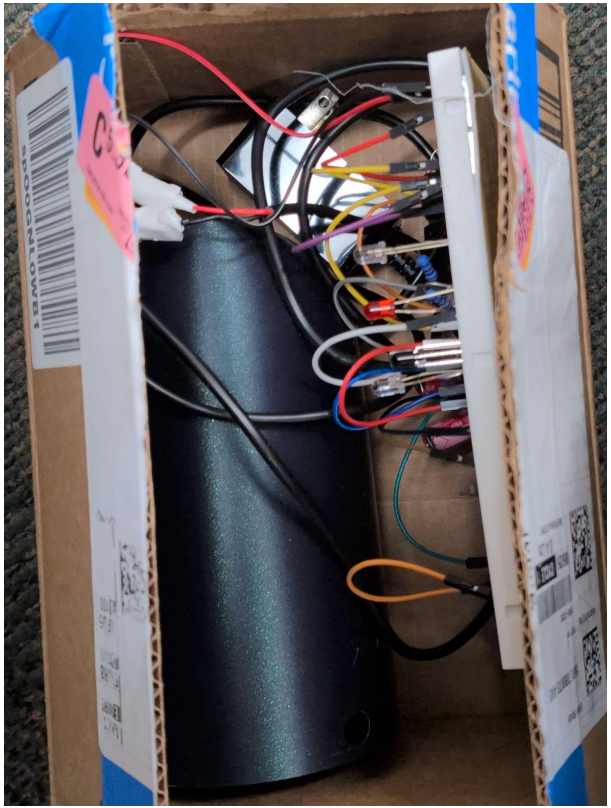
Schematic (updated)

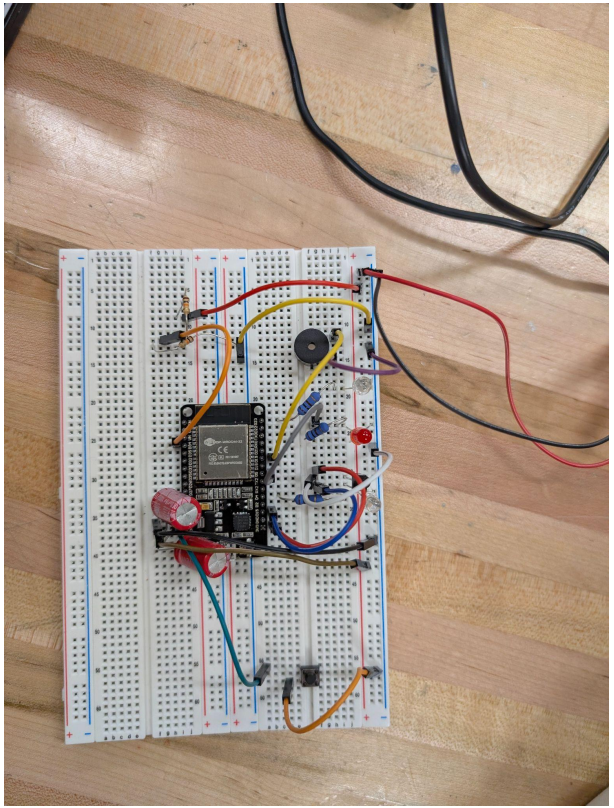


WebUI Testing



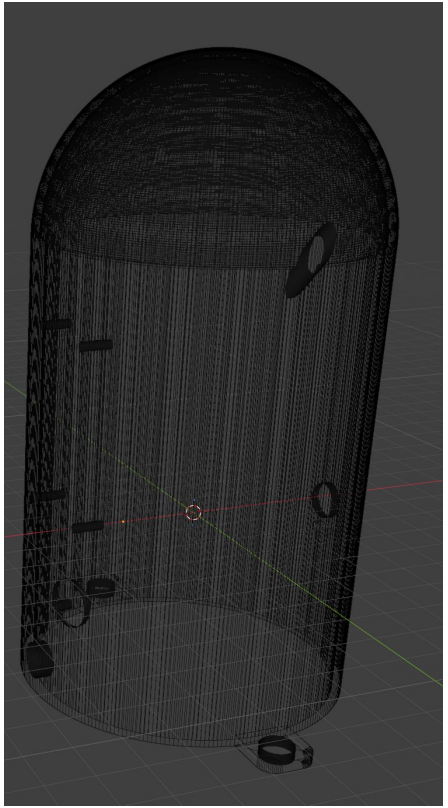
Breadboard testing

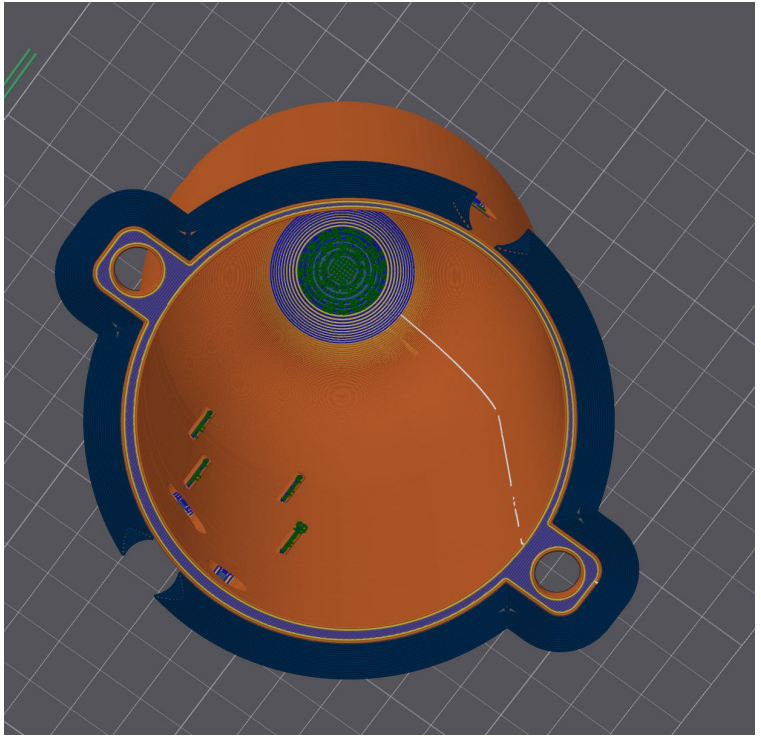




Internal 3D Model & Print







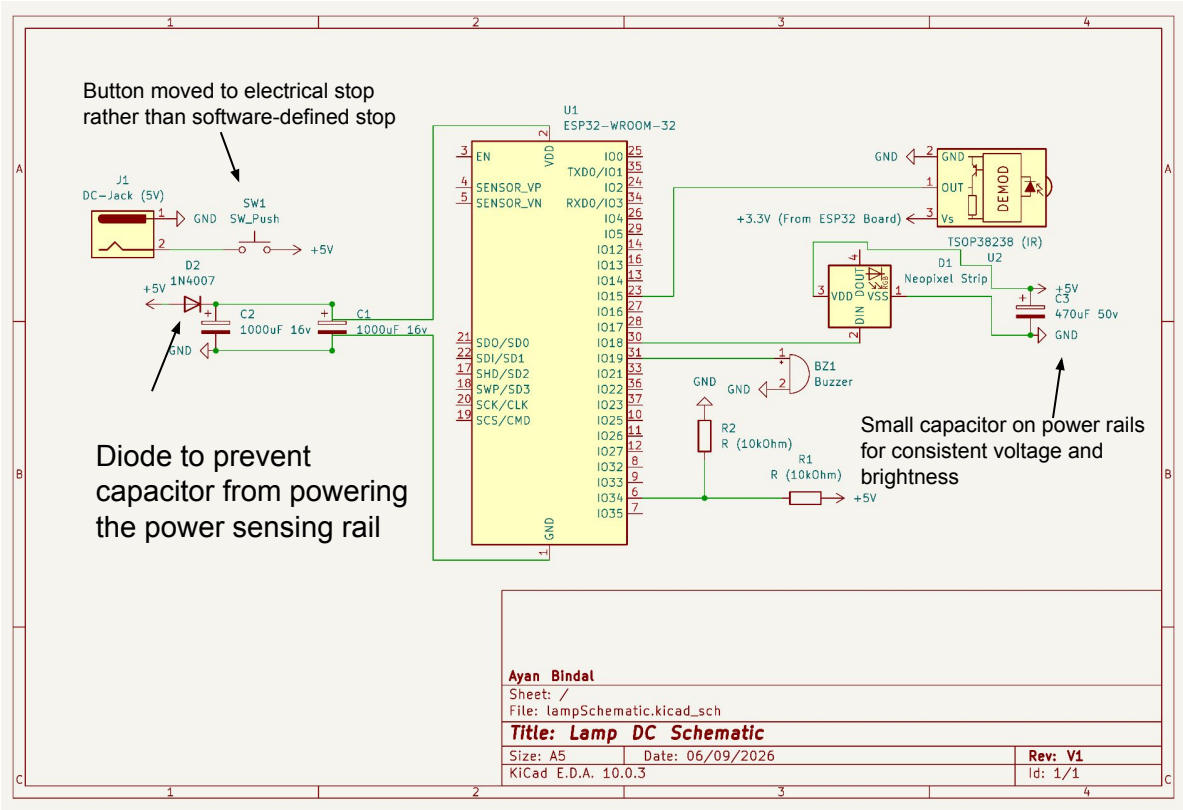
Troubleshooting

- Testing & removal of ambient light sensing
  - Lamp would likely interfere with itself, creating an infinite feedback loop
- Finding IR frequency & pins
  - I had multiple types of IR receivers, finding the correct ones and the pinout was challenging without a part number
- Tweaking power-loss recovery
  - Flash wear leveling
    - Makes sure to only write to the flash when the lamp loses power, helps save on flash write cycles on the ESP32, improves longevity

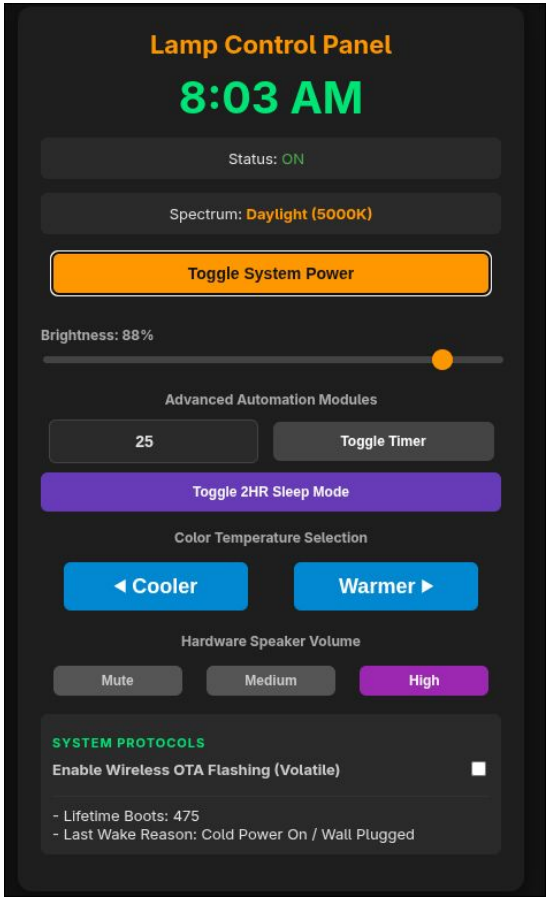
Updated BOM

- Wood PLA for trunk
- Glow PLA for leaves
- Push toggle switch for power
- ESP32 (Rather than the Arduino UNO for WiFi & size)
- Neopixel strip (thicker one with white flex PCB) - 3ft length
- DC to screw terminal block
- DC power adapter (5V/2A)
- IR Receiver (For remote control)
- Buzzer (For system status updates & sound effects)

Final Assembled Circuit



Final WebUI


Lamp Control Panel
8:03 AM
Status: ON
Spectrum: Daylight (5000K)
Toggle System Power
Brightness: 88%
Advanced Automation Modules
25
Toggle Timer
Toggle 2HR Sleep Mode
Color Temperature Selection
◄ Cooler
Warmer ►
Hardware Speaker Volume
Mute
Medium
High
SYSTEM PROTOCOLS
Enable Wireless OTA Flashing (Volatile)
- Lifetime Boots: 475
- Last Wake Reason: Cold Power On / Wall Plugged

Final Result



What Worked Well

- Using Glow-in-the-dark PLA for the lampshade
- Using the ESP32 microcontroller
- Implementing power-loss recovery
- Adding a buzzer for lamp status

Biggest Challenges Faced

- Designing and assembling the actual lamp structure
  - 3D printer size limitations (180x180x180mm)
  - Tolerances
  - Material strength
- Properly sealing the bottom of the lamp
  - Failed screw attempts (poor material & print orientation, shallow threads in trunk)



How Problems Were Solved

- Large tolerances were implemented
  - This led to large gaps, but most were not visible during normal operation
- Metal screws & a mix of metal & 3D printer washers were used to seal the bottom rather than 3D printed screws
  - Needed to put brass heat-set inserts into shallow threads of the trunk to make metal screws work
- Lots of iterations & troubleshooting for any and all other problems
  - Ex. Button sizing & placement, bottom cap tolerances, 3D printing supports tuning

What I Learned About Electronics

I learned that you should always make sure to plan ahead, especially keeping wires in mind rather than just fitting parts into an enclosure. Since I typically work with PCBs, this is less of a concern, but is made much more significant with wires going everywhere.

What I Would Improve

- Remove (or partially remove) the magnets embedded in the lampshade & trunk
  - Do not significantly increase the grip between the two objects
- Design the entire lamp as one piece rather than having an electronics enclosure with a tree shell
  - Makes a lot more room to work with, simpler overall design, less tolerances to worry about, better power, IR, and button placement
- Solder connections between LED strips
  - Can be somewhat fragile